Load Balancing Practical on macOS

Step 1: Open Terminal

Press:

Command (⌘) + Space → type "Terminal"

Step 2: Check Python installation

macOS usually has Python, but verify:

python3 --version

If not installed:

- Install via Homebrew:

$ brew install python

Step 3: Create project folder

mkdir ~/Desktop/load-balancing-lab
cd ~/Desktop/load-balancing-lab

Step 4: Create server.py file

And copy the below code and paste it. Then save it.

```
from http.server import BaseHTTPRequestHandler, HTTPServer
import sys

PORT = int(sys.argv[1])
INSTANCE = sys.argv[2]

class Handler(BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.end_headers()
        message = f"Hello from instance {INSTANCE}"
        self.wfile.write(message.encode())

server = HTTPServer(("localhost", PORT), Handler)
print(f"Running instance {INSTANCE} on port {PORT}")
server.serve_forever()
```

Step 5: Run multiple servers

Open 3 separate Terminal windows/tabs

Terminal 1:

python3 server.py 8001 1

Terminal 2:

python3 server.py 8002 2

Terminal 3:

python3 server.py 8003 3

Step 6: Test servers

Open browser:

- http://localhost:8001
- http://localhost:8002
- http://localhost:8003

Step 7: Install requests library

pip3 install requests

Step 8: Create load_balancer.py file

Copy and paste the following code. Then save it.

```
import itertools
from http.server import BaseHTTPRequestHandler, HTTPServer
import requests

servers = [
    "http://localhost:8001",
    "http://localhost:8002",
    "http://localhost:8003"
]

cycle = itertools.cycle(servers)

class Handler(BaseHTTPRequestHandler):
    def do_GET(self):
        target = next(cycle)
        response = requests.get(target)
```

```
        self.send_response(200)
        self.end_headers()
        self.wfile.write(response.content)

server = HTTPServer(("localhost", 9000), Handler)
print("Load balancer running on port 9000")
server.serve_forever()
```

Step 9: Run load balancer

Open a new Terminal:

python3 load_balancer.py

Step 10: Test load balancing

Open: http://localhost:9000

Refresh multiple times → responses rotate.

Step 11: Simulate failure

Stop one server: CTRL + C
You'll notice:

- Errors or delays

Step 12: Update for fault tolerance

Edit file: load_balancer.py

Replace do_GET method with the following code.

```
class Handler(BaseHTTPRequestHandler):
    def do_GET(self):
        for _ in range(len(servers)):
            target = next(cycle)
            try:
                response = requests.get(target, timeout=1)

                self.send_response(200)
                self.end_headers()
                self.wfile.write(response.content)
                return

            except:
                continue

        self.send_response(500)
```

```
        self.end_headers()
        self.wfile.write(b"All servers are down")
```

Restart load balancer

CTRL + C
python3 load_balancer.py

macOS Tips (important)

- Always use python3 (not python)
- Always use pip3
- If permission error: pip3 install --user requests